

Consistency Questionnaire Results

Q1. Missing physiological attributes

■ No inconsistencies found.

Q2. Multiple HR per ActorState

■ No inconsistencies found.

Q3. Multiple HRV per ActorState

■ No inconsistencies found.

Q4. Multiple RR per ActorState

■ No inconsistencies found.

Q5. Multiple SpO2 per ActorState

■ No inconsistencies found.

Q6. Multiple Drowsiness per ActorState

■ No inconsistencies found.

Q7. Missing high-level attributes

■ No inconsistencies found.

Q8. Multiple Fatigue per ActorState

■ No inconsistencies found.

Q9. Multiple Attention per ActorState

■ No inconsistencies found.

Q10. Multiple Unresponsiveness per ActorState

■ No inconsistencies found.

Q11. Multiple EyeState per ActorState

■ No inconsistencies found.

Q12. Multiple MouthState per ActorState

■ No inconsistencies found.

Q13. Multiple Age or Sex per ActorState

■ No inconsistencies found.

Q14. Multiple Accessories/Demographic/FaceCharacteristics per ActorState

■ 7 inconsistencies found.

```
Running Q14. Multiple Accessories/Demographic/FaceCharacteristics per ActorState...
■ Inconsistencies found (5 rows)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_dc10f2289aa545cf914
eaa7db1ef3861'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_78f12e15020a44598bb
7be45bf7cdb26'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_dfec112258284d9e993
9b1e727a8affc'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_4109c3363f7e4454ae7
a893574d0c24c'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_1995c41469984296beb
292e301833239'),)
```

Q15-A. HR instances classified into more than one HR-level class

■ No inconsistencies found.

Q15-B. HRV instances classified into more than one HRV-level class

■ No inconsistencies found.

Q15-C. RR instances classified into more than one RR-level class

■ No inconsistencies found.

Q15-D. SpO2 instances classified into more than one SpO2-level class

■ No inconsistencies found.

Q15-E. Drowsiness instances classified into more than one Drowsiness-level class

■ No inconsistencies found.

Q16-A. HR without any HR-level classification

■ No inconsistencies found.

Q16-B. HRV without any HR-level classification

■ No inconsistencies found.

Q16-C. RR without any RR-level classification

■ No inconsistencies found.

Q16-D. SpO2 without any SpO2-level classification

■ No inconsistencies found.

Q16-E. Drowsiness without any Drowsiness-level classification

■ No inconsistencies found.

Q17-A. HR instances with >1 HR level

■ No inconsistencies found.

Q17-B. HRV instances with >1 HRV level

■ No inconsistencies found.

Q17-C. RR instances with >1 RR level

■ No inconsistencies found.

Q17-D. SpO2 instances with >1 SpO2 level

■ No inconsistencies found.

Q17-E. Drowsiness instances with >1 Drowsiness level

■ No inconsistencies found.

Q18. Violation "Sleep=> Inattentive & Unresponsive"

■ No inconsistencies found.

Q19. Violation "Drowsy & attentive=> at-risk unresponsiveness"

■ No inconsistencies found.

Q20. Violation "Sleep or unresponsiveness => eyes closed & mouth open"

■ No inconsistencies found.

Q21. Is any state missing its threshold profile?

■ No inconsistencies found.

Q22. Are there any threshold profiles not used by an Actor State?

■ 7 inconsistencies found.

```
Running Q22. Are there any threshold profiles not used by an Actor State?...
■ Inconsistencies found (17 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
```

Q23. Detect future categories accidentally left unmapped

■ No inconsistencies found.

Q24. Show if new data falls outside modeled thresholds.

■ No inconsistencies found.

Q25. Check if new states appear without corresponding GCI or SWRL effects.

■ No inconsistencies found.

Q26. Future adaptability often breaks.

■ 7 inconsistencies found.

```
Running Q26. Future adaptability often breaks....
■ Inconsistencies found (18 rows)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_elderly_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_female_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_middle-aged_male_cold'),)
(rdfLib.term.URIRef('http://test.org/in_cabin_domain#tp_young_female_cold'),)
```

Q27. Severe drowsiness (Level 9) must imply Imminent or Unresponsive

■ No inconsistencies found.

Q28. Moderate drowsiness (Level 5) must be classified as DrowsinessSuspected fatigue

■ No inconsistencies found.

Q29. Low drowsiness (Level 3) must be classified as Awake fatigue

■ No inconsistencies found.

Q30. Low drowsiness (Level 3) must imply Attentive attention

■ No inconsistencies found.

Q31. High drowsiness (Level 7 or 9) must imply Unresponsiveness or DrowsinessSuspected

■ No inconsistencies found.

Q32. Sleep fatigue must have both eye and mouth state defined

■ No inconsistencies found.

Q33. Awake fatigue must be Responsive and Attentive

■ No inconsistencies found.

Q34. High-risk states (Sleep or Unresponsive) must have at least one Label

■ 4 inconsistencies found.

Running Q34. High-risk states (Sleep or Unresponsive) must have at least one Label...

■ Inconsistencies found (2 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_78f12e15020a44598bb7be45bf7cdb26'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#DriverState_1995c41469984296beb292e301833239'),)
```

Q35. Every Label must target at least one ActorState

■ 7 inconsistencies found.

Running Q35. Every Label must target at least one ActorState...

■ Inconsistencies found (5 rows)

```
(rdflib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_1_7'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_16_33'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_24_38'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_28_49'),)
(rdflib.term.URIRef('http://test.org/in_cabin_domain#label_2024_4_15_0_35_56'),)
```